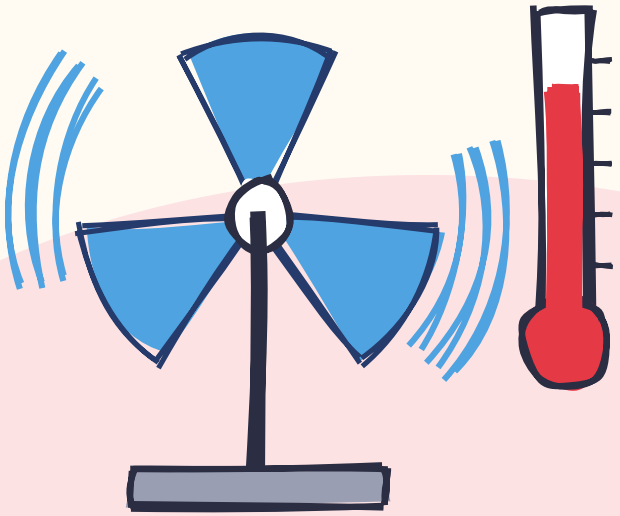
Junior Makers · Class 7

Temperature-Controlled Smart Fan

Hotter room, faster fan. All by itself!

DHT22 sensing | L298N motor driver | Speed control



Did you know?
A fan does not make the air colder. It cools YOU by blowing heat and sweat away from your skin.

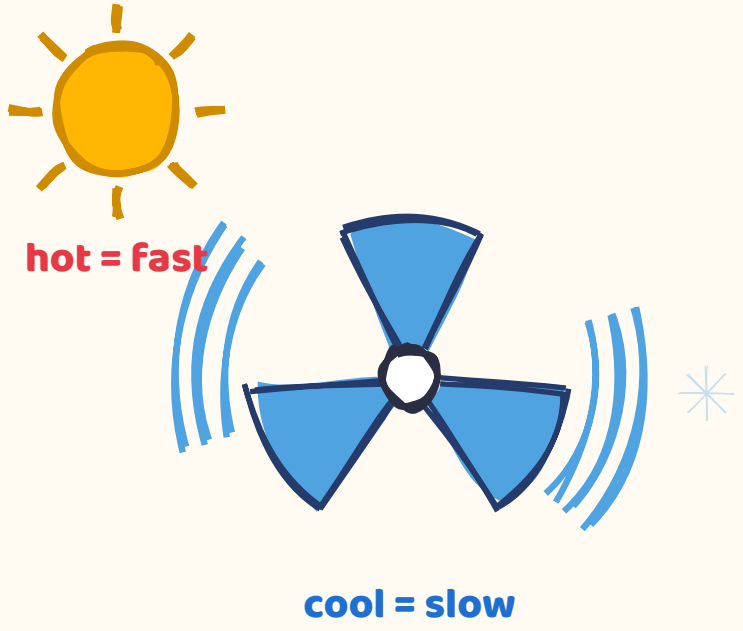
LEARN

Let's think...

Who switches the fan to full speed?

A summer problem

In June the afternoon is **very hot**, so the fan runs at full speed. At night it gets cooler, but nobody turns the fan down. It keeps using **electricity** we do not need.



Set a reminder?

Turn it off?

A fan that reads the temperature!

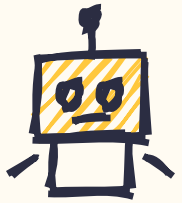
Today: we build a fan that measures the room and picks its **own speed**.

Learning aims

Today I will learn to...

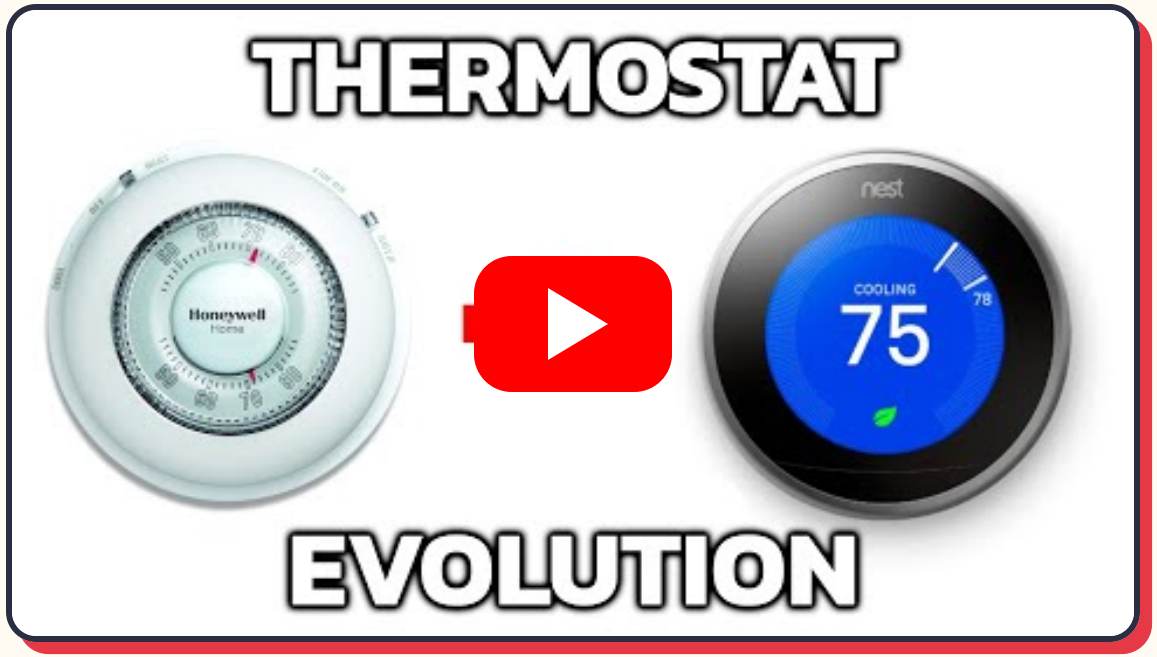
1. Read the temperature with a **DHT22** sensor
2. Explain why a motor needs a **motor driver**
3. Change motor speed with **PWM** (0 to 255)
4. **Map** a temperature range to a speed range
5. Add a **manual mode** with a knob

Keywords
temperature · DC motor · driver · PWM · map · automatic · manual



Watch and think

Video time!



How a Thermostat Works for Beginners

Word of Advice TV · about 5 minutes · youtu.be/Z4T9H-6VPtI

Watch for...

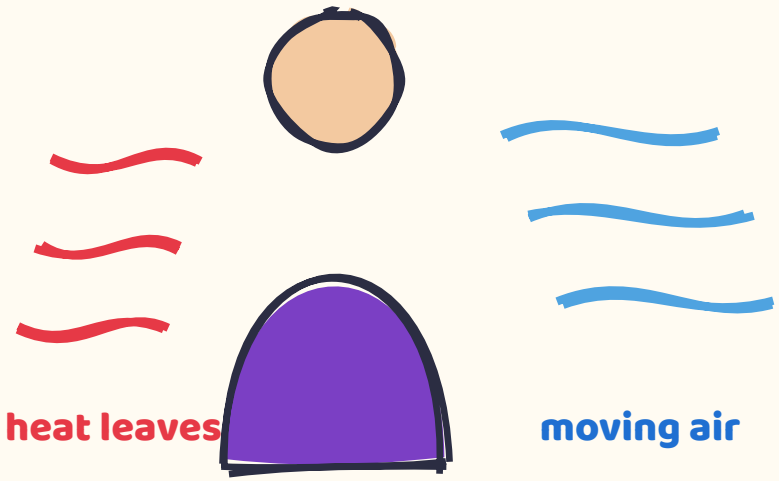
1. What does a **thermostat** measure?
2. What does it switch **on** or **off**?
3. What is the 'set point'?

Scan to watch

After the video: Our smart fan is a thermostat too. What temperature would you choose as its set point, and why?

The science

How does a fan cool us?



Blowing heat away

Our body warms a thin layer of air next to our skin. A fan **blows it away** and brings new air. It also helps **sweat dry**, and drying sweat takes heat with it.

Faster = cooler feeling

More air speed takes more heat away. But a faster motor also uses **more electricity**.

Fun fact

A fan in an EMPTY room does nothing useful. Switch it off when you leave!

Try it

Wet your hand and blow on it. Then blow on a dry hand. Which feels cooler?

Real world

Machines that cool themselves



Laptops
The fan speeds up when you play games and the chip gets hot



Cars
A radiator fan switches on when the engine gets too hot



Greenhouses
Fans and windows open when plants get too warm



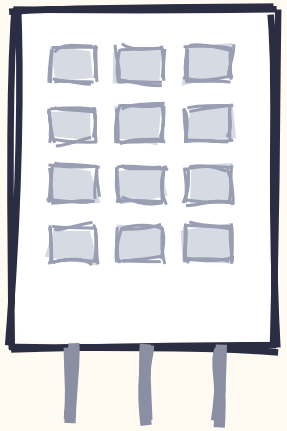
Inverter ACs
Slow down when the room is cool, saving electricity

? Listen to a laptop or a fridge. Can you hear its fan **speed up** or **switch on** by itself?

Same pattern
Sense the temperature, then decide how hard to cool. That is today's project!

Meet the sensor

The DHT22: temperature and humidity



DHT22

Pin	What it does
+	5V
out	Data to pin 2
-	Ground

Fact file

Measures **temperature** in °C and **humidity** (how damp the air is).

Accurate to about **half a degree**.

Needs about **2 seconds** between readings.

Re-use!

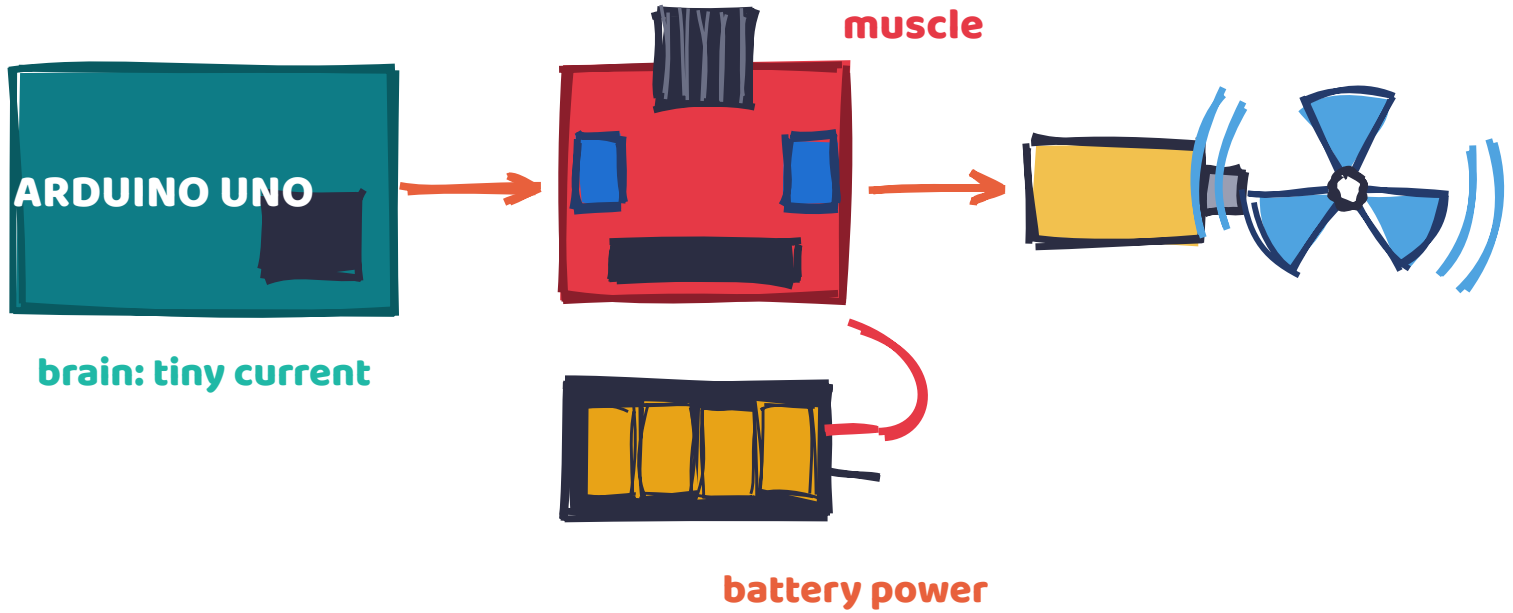
You met this sensor in the **Mini Weather Station**. Today it controls a machine!

Warm it up

Hold the sensor gently between your fingers. The number rises in a few seconds.

Big idea

Why can't the Arduino run the motor?



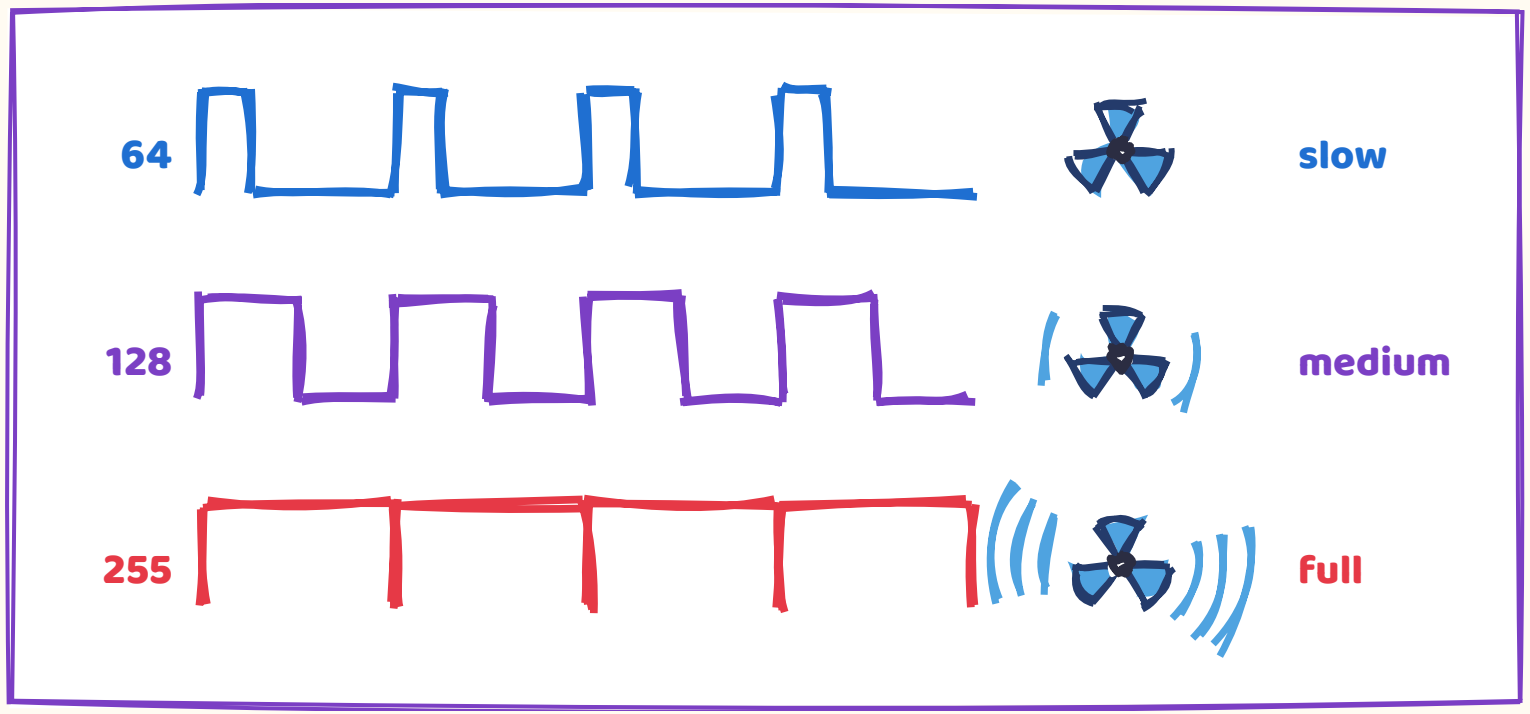
Brain and muscle

An Arduino pin gives only a **tiny** current, like a drinking straw. A motor needs a **lot**, like a garden hose. Connect it straight to a pin and the Arduino could be **damaged**.

The L298N
takes small signals from the Arduino and sends big power from the batteries.

Speed control

PWM: blinking super fast



How it works

The driver switches the motor **ON and OFF** hundreds of times a second. More ON time = more average power = **faster**.

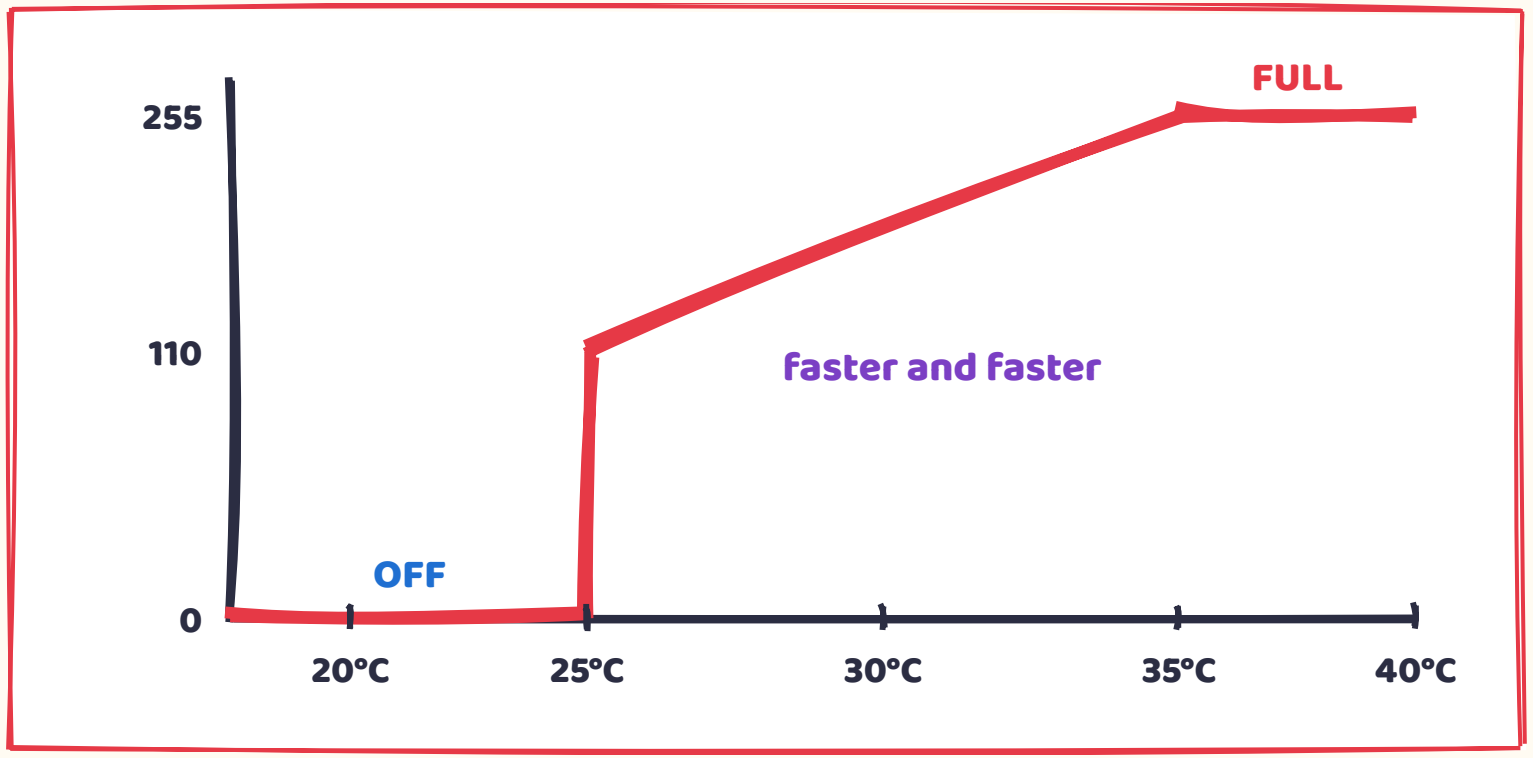
We choose a number from **0** (stop) to **255** (full).

Hmm!

Below about 100 many motors just hum and do not turn. Test yours!

Mapping

Turn temperature into speed



The map block

```
map(temp, 25, 35, 110, 255)
```

25°C becomes speed 110

30°C becomes about 183

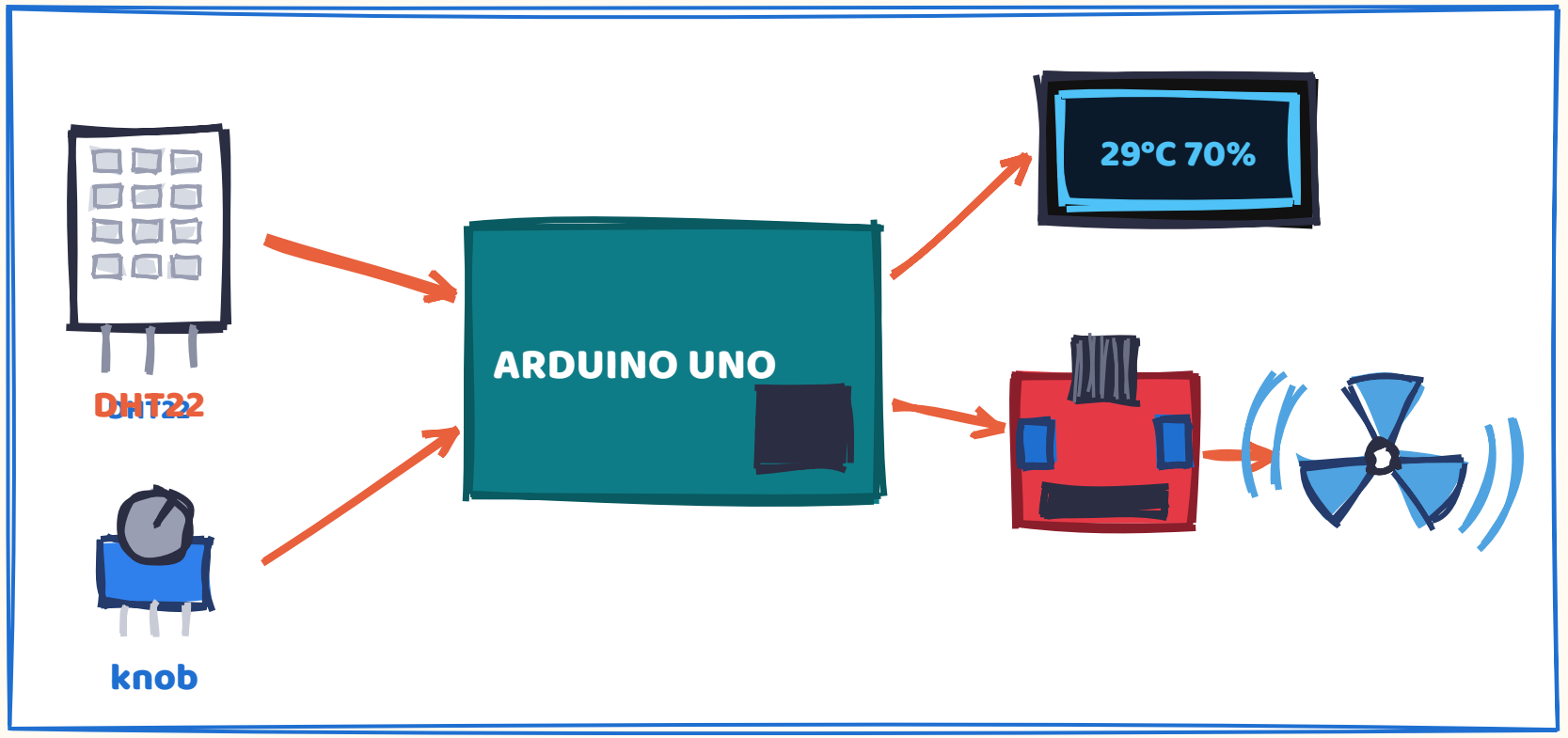
35°C becomes 255

Why start at 110?

Below that our motor would just hum. We skip the useless part!

The plan

Sense, decide, spin



OLED shows

Line 1: temperature
Line 2: fan speed in %
Line 3: AUTO or MANUAL

Knob rule

Knob turned fully left = AUTO. Turn it up = you choose the speed.

Level 2

Add manual mode with the knob

```
set knob to read analog A0
if knob < 50
    AUTO: use the temperature code
    OLED: AUTO
else
    set speed to map(knob, 50, 1023, 110, 255)
    OLED: MANUAL
```

Why have manual mode?

People feel heat **differently**. Grandma may feel cold at 28°C while you feel hot. A good machine lets the **person** take control.

Knob = map again!

The knob gives 0 to 1023. We map it to the speed range, just like temperature.

Design idea

Put the knob where a user can reach it, away from the blade.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Be the smart fan!
2. Our smart fan kit (per group)
3. Wiring the smart fan
4. Auto mode in ArduinoBlocks
5. Does the fan follow the heat?
6. Smart fan challenges

Class game

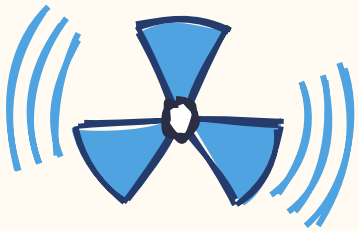
Be the smart fan!

How to play

1. Everyone makes a **paper fan** (fold an A4 sheet).
2. The teacher calls out a temperature: **"22 degrees!"**
3. Below 25: **stop**. 25 to 35: wave **faster** as it rises. Above 35: **full speed!**
4. Wrong speed? You are out!

Then discuss

How did you decide **how fast** to go for 30 degrees?
That middle part is exactly what the Arduino has to work out.



Big idea
You just turned one number (temperature) into another number (speed). That is called MAPPING.

Build it

Our smart fan kit (per group)



Arduino Uno
The brain



DHT22 sensor
Feels the heat



L298N driver
The muscle



DC motor + blade
The fan



10k knob
Manual control



OLED 0.96 in
The display

Also: **4 × AA battery pack** for the motor, breadboard, jumper wires, USB cable, laptop, a card stand for the fan.

Fan safety
Tie back long hair. Keep fingers away from the spinning blade!

Build it

Wiring the smart fan

Part	Connect to
DHT22 + / out / -	5V / pin 2 / GND
OLED SDA / SCL	A4 / A5
Knob middle leg	A0
L298N ENA	pin 9 (remove jumper)
L298N IN1 / IN2	pin 7 / pin 8
L298N OUT1 / OUT2	motor wires
L298N 12V / GND	battery + / -
L298N GND	also Arduino GND!

Common ground!

Join the **battery minus**, the **L298N GND** and the **Arduino GND** together. Without it the signals have nothing to compare against and the motor will not listen.

ENA jumper

Pull off the little black jumper on ENA so pin 9 can control the speed.

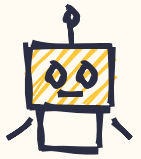
Code it

Auto mode in ArduinoBlocks

```
when Arduino starts
    set up OLED · pin 7 HIGH · pin 8 LOW (spin forward)
forever
    set temp to DHT22 temperature (pin 2)
    if temp < 25
        set speed to 0
    else if temp > 35
        set speed to 255
    else
        set speed to map(temp, 25, 35, 110, 255)
    analog write pin 9 → speed
    OLED: temp and speed
    wait 2000 ms
```

Read it like a story

Cool? **Stop.**
Very hot? **Full speed.**
In between? **Map** it.
Then send the speed to pin 9.



Classroom tip
Our room may already be 28°C! Read your room first, then set the start number 1 degree above it.

Test like an engineer

Does the fan follow the heat?

Test	Expect	Temp	Speed
Room air (no touching)	slow or off	____ °C	____
Hold the DHT22 in your fingers	speeds up	____ °C	____
Let go and wait 1 minute	slows down	____ °C	____
Knob fully left	AUTO on OLED	____ °C	____
Knob half way	MANUAL, medium speed	____ °C	____

Debug checklist

Motor hums only? Raise the lowest speed.
Motor never moves? Check common GND and the ENA jumper.
Temp shows "nan"? Check the DHT22 data wire on pin 2.

Wrong direction? Swap the two motor wires (or swap HIGH and LOW on pins 7 and 8).

Level up!

Smart fan challenges

Choose your challenge



Heat light: add an LED that glows red when it is above 35°C.



Sticky air: also speed up when humidity is above 70%.



Speed bar: draw a bar on the OLED that grows with the speed.



Sleep timer: turn the fan off after 30 minutes in MANUAL mode.

Think like an engineer

If the temperature jumps between 24.9 and 25.1, the fan keeps switching on and off. How could we stop that?

Boss level

Use TWO numbers: switch ON at 26, switch OFF at 24. Engineers call this hysteresis.

Show what you know

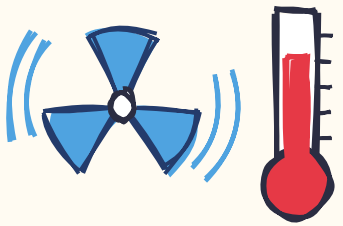
What did we learn?

- [x] A fan cools us by moving heat away from our skin
- [x] The DHT22 measures temperature and humidity
- [x] Motors need a driver: the Arduino is the brain, the L298N is the muscle
- [x] PWM sets the speed from 0 to 255
- [x] map() turns one range of numbers into another

Exit ticket: why must we never connect a motor straight to an Arduino pin?

At home

Find 3 machines at home with a fan inside. Does the fan speed change by itself?



Next class: a Sun-Tracking Solar Panel!